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Dysarthric-speech detection using transfer learning with convolutional neural networks

S R Mani Sekhar*, Gaurav Kashyap, Akshay Bhansali, Andrew Abishek A., Kushan Singh

Department of Information Science & Engineering, Ramaiah Institute of Technology, Bangalore, India

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Abstract

Speech Dysarthria is a disorder in which speech muscles become weak, and it becomes difficult to articulate otherwise linguistically normal speech. This work is based on detection of speech dysarthria and how it can assist physicians, specialists, and doctors in its detection. The proposed work achieves higher accuracies on the TORGO dataset by using a transfer learning based convolutional neural network model (TL-CNN) and by converting the audio samples to Mel-spectrograms. The proposed work TL-CNN achieved better accuracy when compared with other machine learning models.

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Keywords: Dysarthria; Deep learning; Medical technology; Transfer learning; Machine learning

1. Introduction

Patients with speech dysarthria may not realize that this condition is a symptom for multiple other diseases, it can forewarn of a situation such as a stroke or a seizure. Dysarthria can lead to various health issues [1] such as Brain injury, Amyotrophic lateral sclerosis, Brain tumor, Multiple sclerosis, Muscular dystrophy and Parkinson's disease. The proposed work is limited to diagnosing only this symptom and does not aid in diagnosing any cause of such a symptom or any disease.

The objectives of this work are to classify any given speech sample as that from a dysarthric patient or not. The aim is to give a performance metric of accuracy, so as to show that the model determines the sample to be of a dysarthric patient with defined confidence.

The proposed work uses the TORGO database [2] which consists of audio samples from male and female speakers with speech dysarthria and from male and female speakers with no speech impairment. This dataset forms the basis of the proposed solution and model.

* Corresponding author.
E-mail address: manisekharsr@gmail.com (S R Mani Sekhar).

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2. Literature review

In [3], the recorded sound is stored in digital format with two dimensions of frequency and time. The approach combines three things — differences in speech pattern of dysarthric and non-dysarthric persons, different ways of representation of this speech signal and the use of deep learning to learn the pattern. This work uses some additional processing of the input data, feeds it into a CNN model, and trains the model output through a soft-max classifier. Training and testing it on the TORGO [2,4] database, this paper had an accuracy of 68% on detecting if the given audio sample was dysarthric or not.

In [5], the author used neural networks to detect dysarthric speech. They used extended speech features called centroid formants to help differentiate dysarthric and non-dysarthric speech. Later, the work uses experimental dataset that consisted of 200 speeches from 10 Dysarthric patients and 200 speeches from 10 same aged healthy persons. In [6] authors used forced configuration method for speech tempo study, in automatic speech recognition. They used the acoustic measures directly in the tempo adjustments, instead of the time-domain signals.

In [7], the author created an application that selects a feature based on fractal study. They used acoustic & linked articulatory soundtracks of given speech. For classification

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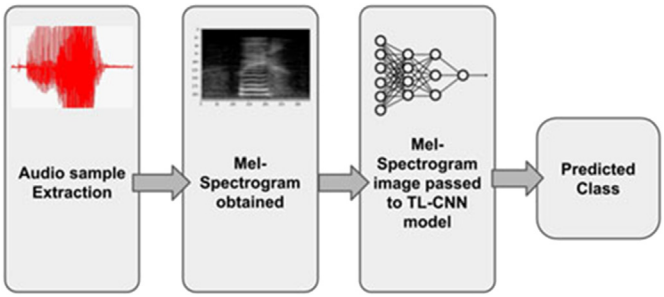


Fig. 1. Flow diagram for the Detection of Speech Dysarthria.

they used Diadocho kinetic test. Whereas in [8] author developed an application for Japanese speech recognition. They used transfer learning (TL) approach for the detection of articulation illness resultant from dyskinetic cerebral palsy. In [9] the author developed an assistive application for dysarthric patients. It used static dimensional vector values to form fixed dimensional score vector representation. They validated the result with UA speech data.

In [10], author has used an empirical and hurst mode as a preprocessing step, later they have integrated the preprocessing method with CNN. The results show that their work performs better when compared with traditional CNN but need more effective model while dealing with the extreme variability speech due to its complex nature.

3. Proposed solution

When an audio sample of a person having dysarthria is extracted, on general observation, it is found that speech from a dysarthric patient is slow, slurred, raspy and has a deeper pitch than a normal person. These characteristics are manifested in the forms of patterns within the audio sample.

This work proposes a system to predict if an audio sample is that from a dysarthric patient or not. Fig. 1 shows the flow diagram for the proposed work.

The audio clip is first converted to a Mel-spectrogram as it visually depicts the logarithmic nature of human hearing perception and can detect the patterns within the audio sample. This Mel-spectrogram image is passed on to our model which performs the prediction of the audio sample.

The detailed process flow of the proposed solution is as follows:

Step 1. Data collection and pre-processing:

i. *Data collection and description:* The TORGO speech dataset [2] is chosen since it contains audio samples of men and women with and without dysarthria, totaling 9094 samples. Fig. 2 illustrates the amplitude waveform of speech samples from that of a dysarthric patient and that of a non-dysarthric patient.

ii. Data Pre-processing: Since Fig. 2 only shows the relationship between amplitude and time and is two dimensional in nature, all the features of the audio clip cannot be represented in one image, hence the audio clip is converted to a Mel-spectrogram to bring out the relationship between amplitude, frequency, and time all in one image. This also normalizes

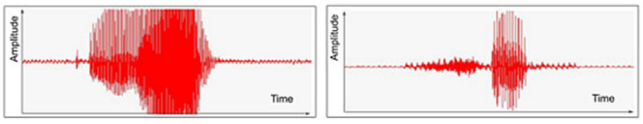


Fig. 2. Amplitude waveform graph of audio samples of a Dysarthric person (left) and that of a non-dysarthric person (right).

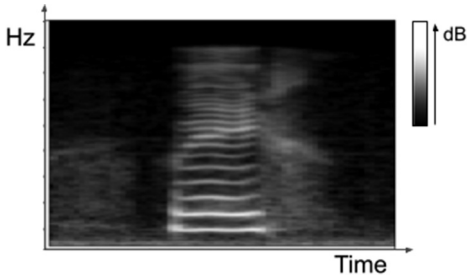


Fig. 3. A Decibel Mel-spectrogram of a speech sample.

the audio spectrum from 0–22 KHz to the Mel-scale, thus improving feature separation as shown in Fig. 3.

iii. Data splitting: The dataset was structured into two classes for the required binary classification. The dataset is split into training and test sets in the ratio 0.7:0.3. The training set consists of 6400 images of which 43.75% are dysarthric samples and 56.25% are non-dysarthric. 10% of the training set is used as the validation set.

Step 2. Modeling:

i. Model design: Since the Mel-spectrogram forms the basis of the input as discussed earlier and since CNNs are best suited for image classification, a CNN based architecture is chosen. Since the objective of this work is to show how a transfer learning aided CNN model (TL-CNN) can perform better on the TORGO dataset [3], the ResNet50 transfer learning model is chosen as the base to our CNN model. ResNet50 is a popular 50-layer pre-trained CNN designed for image classification. The ResNet50 model is trained on the ImageNet dataset which is a large image database containing more than 14 million images. This model is beneficial since it provides a good architecture and optimized weights for relatively small datasets, like the TORGO dataset. The TL-CNN model is built by appending a hidden layer of 256 units to the output layer of the ResNet50 model, and the 256 unit layer is then connected to a final single neuron layer that is sigmoid activated. The optimizer used is the Adam optimizer, and the loss metrics used are Binary cross-entropy, and Mean Absolute Error (MAE). The architecture diagram of the proposed model is shown in Fig. 4.

ii. Training the model: The model was trained for 10 epochs on the training set, which resulted with a training accuracy of 99.61% and a training binary cross entropy loss of 0.0106. A validation set of 640 samples was also used during the training process. The validation accuracies are shown in Table 1. When the model was trained for 20 epochs the validation accuracy started reducing to 95.56% at 12 epochs, 89.78% at 15 epochs and reduced to 81.31% at the end of 20 epochs while the

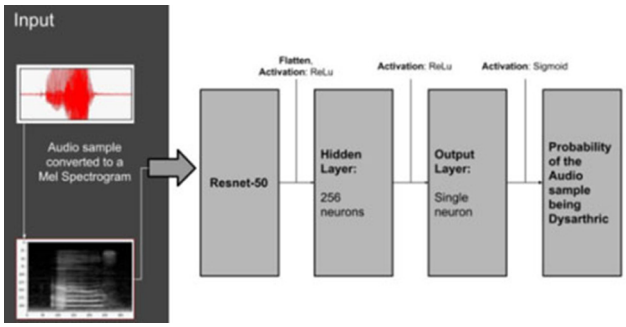


Fig. 4. Architecture Diagram of the Proposed TL-CNN Model.

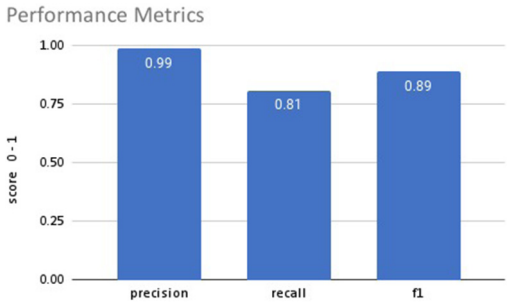


Fig. 5. Precision, recall and F1 score of the TL-CNN.

training accuracy started increasing to 99.68% at 12 epochs, 99.79% at 15 epochs and increased to 99.94% at the end of 20 epochs, indicating over-fitting. Performance is found to dip after 10 epochs and thus 10 epochs are found to be optimal.

Step 3. Evaluation:

i. Testing the model: The proposed model was tested on a new set of 2694 images and achieved a test accuracy of 97.73%. The model achieved a precision, recall and F1 score of 0.99, 0.81 and 0.89, respectively as shown in Fig. 5.

4. Results and discussion

Since the model's precision is high, the model has fewer false positives than false negatives, which is desirable in the testing and detection of speech impairments like dysarthria.

The objective of the proposed model is to perform binary classification, hence the error metric used is Binary cross-entropy, along with MAE, which is used as a comparison metric. Table 1 describes the metrics of the training process of the dysarthria detection model for 10 epochs. The TL-CNN model was evaluated using the following metrics: Accuracy, Binary cross-entropy loss, and MAE. This work also compares the results of the TL-CNN with an Artificial Neural Network (ANN) based on a 2 hidden layer architecture with a final sigmoid activated neuron, a Deep Neural Network (DNN) based on a 5 hidden layer architecture with a final sigmoid activated neuron and a conventional CNN as in [2].

The ANN model achieved a training accuracy of 90.29% which however over-fits past 5 epochs therefore the training accuracy is taken at that point.

The DNN was trained for 15 epochs. Test accuracy was observed to dip and the training accuracy was observed to

Table 1
Obtained accuracy, binary cross-entropy loss, and MAE values and validation accuracy of the TL-CNN model.

Epochs	Training accuracy (%)	Loss	MAE	Validation accuracy (%)
1	87.9	0.717	0.165	70.7
2	97.1	0.0788	0.0441	81.92
3	95.26	0.1223	0.0667	80.4
4	97.22	0.0768	0.0418	79.25
5	98.9	0.0294	0.0159	84.34
6	99.11	0.0242	0.0127	91.04
7	99.39	0.017	0.087	95.84
8	99.36	0.019	0.0095	92.19
9	99.3	0.0219	0.01	96.69
10	99.61	0.0106	0.058	94.19

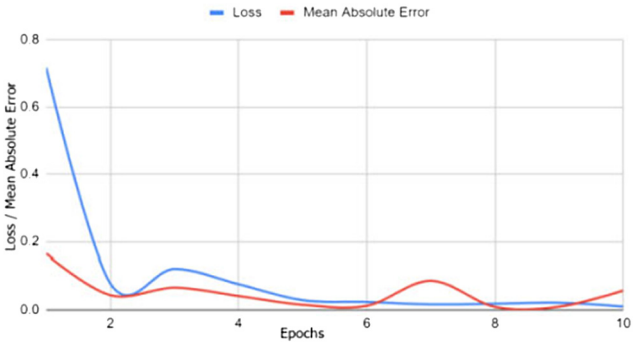


Fig. 6. Loss (Binary Cross-Entropy) and MAE vs number of epochs of TL-CNN.

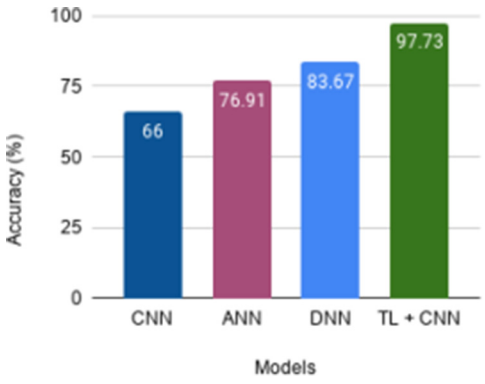


Fig. 7. Comparison of Test Accuracy among CNN, ANN, DNN and TL-CNN.

increase gradually after 8 epochs. This indicated over-fitting; hence the optimal number of epochs was observed to be 8. The test accuracy at the optimal 8 epochs was found to be 83.67% while the test accuracy at 15 epochs was found to be 72.41%.

Fig. 6 shows the progression of Binary cross-entropy loss and MAE against the number of training cycles (epochs) of the TL-CNN.

Fig. 7 shows the difference among the best test accuracies of a conventional CNN [2], ANN, DNN and TL-CNN. The result shows that TL-CNN achieves an accuracy of 97.73% meanwhile CNN is 66%, ANN is 76.91%, and DNN is 83.67%. The proposed work is further validated using the

precision, recall and the F1 parameters. Its precision score is 0.99, recall score is 0.81 and F1 score is 0.89.

5. Conclusion

This work achieved an accuracy of 97.73% using the proposed TL-CNN based method to detect the presence of dysarthric speech in a sound sample, elucidating that the TL-CNN model performs best on the TORGO dataset.

CRediT authorship contribution statement

S R Mani Sekhar: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Gaurav Kashyap:** Conceptualization, Formal analysis, Writing – original draft. **Akshay Bhansali:** Conceptualization, Data curation, Writing – original draft. **Andrew Abishek A.:** Conceptualization, Data curation, Formal analysis, Writing – original draft. **Kushan Singh:** Conceptualization, Formal analysis, Data curation, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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